Preference over Choices

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Abstract

Bringing about the same outcome can earn praise, blame, or neither, depending on what was given up and at what cost. I develop a decision-theoretic model of these judgments, which underlie how people reward and punish one another. From an observer's ranking of another person's choices (outcome–menu pairs), I separate her valuation of outcomes from her praise or blame for choosing them. She compares his actual choice with the one she expects and credits him for the sacrifice she expects him to make. These expectations depend on what she thinks ought to be chosen and what she believes he values. Hence, the more difficult she believes doing the right thing is for him, the more she praises success and the less she blames failure. The model explains why observers who agree on what ought to be chosen can disagree about praise and blame. By deriving these judgments from an observer's preferences, the model clarifies the restrictions built into familiar measures of kindness in theories of reciprocity and accommodates judgments those measures exclude.

Keywords: Preferences over choices; moral judgment; praise and blame; sacrifice; reciprocity

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1. Introduction

The same outcome can come from a good choice or a bad one, depending on what was forgone in choosing it. Consider a judge assessing a defendant for a harm that he caused. Suppose first that he caused it by pure accident, through no choice of his. The judge neither blames nor praises him: there is no choice for her to judge. This case is a natural point of comparison for the cases in which he did make a choice. Had he chosen to cause the same harm when he could have avoided it, she would judge his choice worse than the accident. Had he caused it to prevent a greater harm, choosing the lesser of two evils, she would judge his choice better; the law treats such conduct as justified (Model Penal Code § 3.02).[1]

Not every choice, however, is judged differently from having no choice. What also matters is how difficult the choice was. Suppose he caused the harm because he was threatened with death if he refused. The law excuses him, because a "person of reasonable firmness" in his situation could not have resisted the threat (Model Penal Code § 2.09); it treats his choice exactly as if he had had no choice. Likewise, choosing the lesser evil earns praise only if it required a sacrifice. If preventing the greater harm also meant saving his own skin, his choice is still justified but easy, morally neutral, and no more praiseworthy than the accident. Moral philosophers have defended the principle at work: the more difficult it is to act well, the more praiseworthy one is for succeeding and the less blameworthy for failing (Nelkin, 2016). Difficulty is also where judgments diverge. Suppose refusing the threat would have put him at risk, but not of death. How much risk can reasonably be asked of a person? Two judges who agree that causing the harm was wrong may disagree about whether he deserves blame, because they believe differently about how costly refusing would have been for him.

A judgment of a choice therefore has two parts. The alternatives the defendant gave up determine its direction, toward praise or toward blame relative to having had no choice. The difficulty of doing the right thing determines whether, and how much, his choice actually earns praise or blame. This paper provides an axiomatic model of such judgments. An evaluator (she) has a preference $\succsim$ over the choices of a decision maker (DM, he), where each choice is a pair (x, A) of an outcome x and the menu A from which the DM chose it. The evaluator takes each menu as exogenously given rather than designed by the DM, as a court takes as given the circumstances the defendant faced.[2] I ask two questions. First, what structure must the preference $\succsim$ have if she is both a consequentialist, who cares about

[1] The Model Penal Code, published by the American Law Institute in 1962, serves as a standard reference for the principles of criminal liability in many U.S. states (Wechsler, 1968; Robinson and Dubber, 2007).

[2] If the defendant deliberately brought about the situation he faced, the law judges the earlier choice that created it (see Model Penal Code §§ 2.09(2), 3.02(2)). Treating menus as exogenous conforms to this practice, while abstracting away from any dynamics.

the outcomes of choices, and a judge, who evaluates the quality of each choice against the alternatives the DM gave up? Second, how does her preference capture the difficulty she perceives in the menu the DM faces? I show that we can address these questions using standard techniques of decision theory.[3]

My main results are two representation theorems. The first (Theorem 1) characterizes the general class of such preferences: what any such ranking must look like. The second result (Theorem 2) adds behavioral content: it identifies the evaluator's belief about how much the DM values each alternative, which determines the difficulty she perceives in the menu he faces.

First, any evaluator of the above nature assesses each choice according to

$$U(x, A) = U(x, \{x\}) + v(x) - v(r(A)),$$

where each outcome x is a lottery over some sure alternatives and menu A is a set of those lotteries. The term $U(x, \{x\})$ is the value of x when it comes about without a choice, as in the accident, and it represents her ranking of outcomes. This is the consequentialist's part of the evaluator.[4] The judge's part is the function $Q(x, A) = v(x) - v(r(A))$, where v is a von Neumann–Morgenstern (vNM) utility function over lotteries and $r(A)$ is a mixture of the options on menu A. This captures the quality of choosing x from A. The function v represents her ideal ranking, the one by which the judge wants the DM to choose. The quality is measured against the benchmark $r(A)$: choosing $r(A)$ from menu A would be judged exactly as if the DM had been forced into it. Since $U(x, A) - U(x, \{x\}) = Q(x, A)$, choosing x from A is better than having had no choice exactly when the ideal ranking puts x above the benchmark. I call this difference praise when it is positive and blame when it is negative.

The axioms that deliver this result are twofold. I first use a natural extension of the standard axioms of expected utility (EU) theory to additively separate the consequentialist from the judge: $\succsim$ is complete, transitive, continuous, and independent. The framework thus relies on lotteries over choices, allowing us to characterize a ranking of choices that is independent of the ranking of outcomes. Consider $(\frac{1}{2}x + \frac{1}{2}y, \frac{1}{2}A + \frac{1}{2}\{y\})$ and $(\frac{1}{2}y + \frac{1}{2}x, \frac{1}{2}B + \frac{1}{2}\{x\})$. Both yield the same distribution of outcomes. In the first, x is chosen from A and y is imposed; in the second, y is chosen from B and x is imposed. Under the EU axioms, it

[3] As in Nelkin (2016), praise and blame in this paper are meant in the accountability sense, not as an inference from a choice about a person's character. Moreover, I model difficulty as the sacrifice of one's interests, although it can also mean the effort an action requires.

[4] I include the consequentialist in the evaluator to reflect that people respond to outcomes even when no one chose them: voters, for instance, punish governments for bad weather beyond their control (Cole et al., 2012). A pure judge is the special case in which $U(x, \{x\})$ is constant in x.

reduces to comparing $Q(x, A)$ with $Q(y, B)$. The next axiom, called Relativity, states that for each menu A, we have $(x, A) \succsim (x, \{x\})$ and $(y, \{y\}) \succsim (y, A)$ for some $x, y \in A$. That is, every menu contains a choice the evaluator judges at least as good as having had no choice and one she judges at least as bad. Consequently, the benchmark $r(A)$ lies between the least and most ideal options in menu A. This is the key structure for the judge, requiring her to be impartial about menus: the menu alone never earns the DM praise or blame.[5]

The second representation adapts the model of temptation and self-control (Gul and Pesendorfer, 2001), which identifies two conflicting rankings of outcomes from a ranking of menus, to the present framework. It replaces the abstract benchmark $r(A)$ with a specific judgment rule by identifying one additional vNM utility function τ, which I interpret as the evaluator's belief about how much the DM values each alternative. While conforming to the structure of the first representation, the judge's part takes the form

$$Q(x, A) = v(x) - v(\hat{x}(A)) + c(A),$$

where $\hat{x}(A)$ maximizes $v + \tau$ over A, and $c(A) = \max_{z \in A} \tau(z) - \tau(\hat{x}(A))$.[6] I say the option $\hat{x}(A)$ is the choice the evaluator expects the DM to make: it balances her ideal ranking v against what she believes he values, τ. The term $c(A) \geq 0$ is the cost she expects him to bear in choosing $\hat{x}(A)$ instead of the option he values most. Praise and blame therefore have two separate sources. The first, $v(x) - v(\hat{x}(A))$, is surprise: it is positive when the DM does better than she expected, by her ideal ranking, and negative when he does worse. The second, $c(A)$, is expected cost: it raises the quality of every choice from a menu in which she expects the DM's choice to be costly to him. An option added to a menu therefore changes the judgment of other choices through the expected choice and its expected cost.[7]

To see how the representation works, return to the defendant under a threat. By the judge's ideal ranking, refusing to cause harm is the right choice. If she believes that refusing costs him nothing, either because the threat carries no real risk or refusing also saves his own skin, she expects him to refuse; nothing about that choice is surprising or difficult. Refusing then earns no praise, and causing the harm is fully blamed. If refusing is costly but somewhat tolerable, she still expects him to refuse, and refusing earns praise for bearing that cost; giving in is blamed, but less than when refusing cost nothing. If he is under

[5] This reflects criminal law, in which liability must rest on the defendant's conduct rather than on his circumstances alone, and every situation leaves room for blameless conduct (see Model Penal Code §§ 2.01(1), 2.02(2)(d), 2.09(1)).

[6] Theorem 2 identifies a unique τ, under a condition analogous to that of Gul and Pesendorfer (2001).

[7] Suppose the benchmark in the first representation is interpreted as the choice the evaluator expects the DM to make. Then $U(x, A)$ resembles the expectations-based reference-dependent utility of Kőszegi and Rabin (2006). Apart from loss aversion, the judge's part takes the form of their gain-loss utility, and praise and blame measure surprise alone. In the second representation, the benchmark is not the expected choice.

a threat of death, however, she expects him to give in, which costs him nothing. Giving in is then judged as if he had had no choice, and refusing would earn praise for exceeding her expectation. Whether a given threat makes refusing tolerable or prohibitive depends on how strongly τ favors causing the harm, compared with how strongly v favors refusing. Hence, two judges with the same ideal ranking can disagree in the hard case. Proposition 1 formalizes these three cases.

The second theorem not only captures the desired moral principle but also illustrates a technical connection between preferences over choices and preferences over menus. In Gul and Pesendorfer (2001)'s model of temptation, a DM chooses a menu in anticipation of his later choice from it, and his preference over menus is represented by a function of sets. In this paper, the central object is also a function of sets: the benchmark value $R(A) \coloneqq v(r(A))$. By the first theorem, R is an affine function, as is their utility over menus. In the second representation, R satisfies their key axiom, Set Betweenness: $R(A) \geq R(B)$ implies $R(A) \geq R(A \cup B) \geq R(B)$. The corresponding axiom for the evaluator's preference is called Choice Betweenness: $(x, A) \gtrsim (x, B)$ implies $(x, A) \gtrsim (x, A \cup B) \gtrsim (x, B)$. Under the first representation, $(x, A) \gtrsim (x, B)$ holds exactly when $R(A) \leq R(B)$. Choice Betweenness is therefore Set Betweenness for R, with one difference: since (x, A) and (x, B) are choices only when x belongs to both menus, it restricts the benchmarks only of menus that share an option. I show that, in the space of lotteries, this restriction is without loss: Choice Betweenness still delivers Set Betweenness for R. More generally, this result demonstrates that the axiomatic structure of menu preferences can be carried over to outcome-menu pairs to explore more refined judgment rules. Given the large literature that followed Gul and Pesendorfer (2001), it opens the door to many ways of modeling the DM's difficulty.

Conceptually, however, the two frameworks study different objects. The literature on menu preferences is indeed built on the idea that a DM values his own choices (see Lipman and Pesendorfer, 2013, for a survey). Yet modeling a DM's preference over his own choices directly would depart from the principle of revealed preference.[8] Abstracting from problems of self-control, this paper instead takes an observer's ranking of another person's choices as the primitive.

This extension of the domain is one of the paper's main contributions. It provides a behavioral foundation for analyzing rewards, punishment, and regulation when people's responses depend on the quality of choices as well as on their outcomes. The evaluator's preference has economic meaning. Beyond the running example of criminal law, the way choices are judged shapes the incentives of those who make them: voters reelect or remove

[8] Dekel et al. (2009) make this point about extending preferences over menus to outcome-menu pairs. If a DM facing menu $A = \{x, y, z\}$ would choose x, no choice can reveal whether he ranks (y, A) above (z, A).

politicians, parents praise or discipline their children, and consumers punish firms' unfair pricing decisions (Kahneman et al., 1986).

Theories of intention-based reciprocity build such judgments into strategic interaction: a player rewards actions of the other player that are judged kind and punishes those judged unkind. Experiments and surveys show that these responses depend on the other player's menu (Falk et al., 2003, 2008) and on the sacrifice that an action involves (see Jang and Sohn, 2024). These theories, however, specify the judgment rule in advance, as an ingredient of the game.[9] Abstracting from strategic interaction, this paper instead derives the rule from the evaluator's ranking of choices.[10]

The representations also clarify what these rules assume and what they omit. A kindness measure based on who gets more than whom (Falk and Fischbacher, 2006) conflates the evaluation of outcomes with the judgment of choices, which the first representation separates. The widely used midpoint rule (Rabin, 1993; Dufwenberg and Kirchsteiger, 2004; Jang and Sohn, 2024) is induced in the second representation by a unique belief about the DM (Proposition 2).[11] Specifying this rule in advance thus amounts to specifying the evaluator's belief in advance. The second representation also captures the principle of Nelkin (2016): the costlier acting well is expected to be, (i) the more the DM is praised for succeeding and (ii) the less he is blamed for failing. Jang and Sohn (2024), who incorporate sacrifice, capture only (i). Section 4 presents these comparisons in detail.

The remainder of the paper is organized as follows. Section 2 introduces the framework and the axioms and presents the two representation theorems. Section 3 studies how the second representation assigns praise and blame. Section 4 compares the two representations with the measures of kindness in theories of reciprocity. Section 5 derives the theorems from the axioms. Section 6 distinguishes judging a choice from inferring the DM's character from it, discusses other related literature, and outlines extensions. Section 7 concludes. Omitted proofs are in the Appendix.

[9] Segal and Sobel (2007) axiomatize preferences for reciprocity, but their axioms place no restriction on how a player judges the behavior of opponents.

[10] Unlike the objects of standard choice data, a choice is not something the evaluator can choose and implement. Relating her preference to observable rewards and punishments requires embedding the representations in a strategic interaction, such as a model of reciprocity, which I leave for future work.

[11] The midpoint rule measures kindness against the average of the highest and lowest payoffs that one player could have given the other. Dufwenberg and Kirchsteiger (2004, footnote 7) write, "We see no deep justification for picking the average (rather than some other intermediate value), except that the choice is simple and does not affect the qualitative performance of the theory." The specification has since been debated (Dufwenberg and Kirchsteiger, 2019; Isoni and Sugden, 2019), and experiments find that the distance to the midpoint is a poor measure of kindness (Hinz et al., 2024).

2. Model and Representation Theorems

This section introduces the framework, states the axioms, and presents the two representation theorems. Section 5 derives the theorems from the axioms.

2.1. Choices

Let Z be a finite set of sure alternatives, and let X be the set of lotteries on Z, endowed with a metric d that generates the weak topology. Elements $x, y, z \in X$ are called outcomes or options. Let $\mathbb{M}$ be the set of nonempty compact subsets of X, called menus, endowed with the Hausdorff metric.[12] For $A \in \mathbb{M}$, $\text{conv}(A)$ denotes the convex hull of A, which can be interpreted as the menu that offers every randomization of the options in A.

A *choice* is a pair (x, A) with $x \in A \in \mathbb{M}$, meaning that the DM chose outcome x from menu A. I write (x, A) only when $x \in A$. Let $\mathbb{C} := \{(x, A) : x \in A \in \mathbb{M}\}$ be the set of choices, endowed with the product topology. The choice $(x, \{x\})$ describes the case in which x is imposed on the DM, who forgoes no alternative.

Choices can be mixed. For $A, B \in \mathbb{M}$ and $\lambda \in [0, 1]$, let $\lambda A + (1-\lambda)B := \{\lambda x + (1-\lambda)y : x \in A,\ y \in B\}$, and let $\lambda(x, A) + (1-\lambda)(y, B) := \big(\lambda x + (1-\lambda)y,\ \lambda A + (1-\lambda)B\big)$. This mixture describes a lottery over choices: with probability λ, the DM faces A and chooses x, and with probability $1-\lambda$, he faces B and chooses y. Its outcome, $\lambda x + (1-\lambda)y$, is the lottery he induces, and its menu, $\lambda A + (1-\lambda)B$, is the set of lotteries he could have induced by choosing differently from each menu.[13] By identifying the lottery over choices with this pair, the domain builds in a reduction of compound lotteries that applies to outcomes and menus alike (Samuelson, 1952).[14]

The primitive is a binary relation $\succsim$ on $\mathbb{C}$: the evaluator's preference over the DM's choices. As usual, $\succ$ and $\sim$ denote its asymmetric and symmetric parts.

2.2. Axioms

The first three axioms are natural extensions of the standard vNM axioms.

[12] For $A, B \in \mathbb{M}$, $d_H(A, B) := \max\{\max_{x \in A} \min_{y \in B} d(x, y),\ \max_{y \in B} \min_{x \in A} d(x, y)\}$.

[13] Alternatively, one might interpret λ as relative frequency: menu A was given λ times in which case, the DM chose x, and menu B the remaining $1-\lambda$ times, in which case he chose y. Together, these extend Ok and Tserenjigmid (2022)'s interpretations of stochastic choice—originally defined for a single deterministic menu—to the space of lotteries where both choices and menus are stochastic.

[14] Equivalently, the primitive could be a preference over lotteries over choices, $\Delta(\mathbb{C})$ with this reduction imposed as an axiom. The lottery $\lambda \circ (x, A) + (1-\lambda) \circ (y, B) \in \Delta(\mathbb{C})$ that yields the choice (x, A) with probability λ and (y, B) with probability $1-\lambda$ would be treated as equivalent to the reduced choice $(\lambda x + (1-\lambda)y, \lambda A + (1-\lambda)B) \in \mathbb{C}$. Gul and Pesendorfer (2001) and Dekel et al. (2001, 2007) use an analogous construction of lotteries over menus to motivate the Independence axiom for preferences over menus.

Axiom 1 (Weak Order)**.** $\succsim$ *is complete and transitive.*

Axiom 2 (Independence)**.** *For all* $(x, A), (y, B), (z, C) \in \mathbb{C}$ *and* $\lambda \in (0, 1)$*, if* $(x, A) \succ (y, B)$*, then* $\lambda(x, A) + (1 - \lambda)(z, C) \succ \lambda(y, B) + (1 - \lambda)(z, C)$*.*

Axiom 3 (Continuity)**.** *For every* $(y, B) \in \mathbb{C}$*, the sets* $\{(x, A) \in \mathbb{C} : (x, A) \succsim (y, B)\}$ *and* $\{(x, A) \in \mathbb{C} : (y, B) \precsim (x, A)\}$ *are closed.*

Axiom 4 (Relativity)**.** *Each* $A \in \mathbb{M}$ *satisfies* $(x, A) \succsim (x, \{x\})$ *and* $(y, \{y\}) \succsim (y, A)$ *for some* $x, y \in A$*.*

Relativity states that every menu offers a choice that the evaluator judges no worse than having its outcome imposed, and a choice that she judges no better. The name reflects the idea that the evaluator, in her role as a judge, assesses each choice relative to the available alternatives.

2.3. Representation 1: The Preference over Choices

A function $f : X \to \mathbb{R}$ is *affine* if $f(\lambda x + (1-\lambda)y) = \lambda f(x) + (1-\lambda)f(y)$ for all $x, y \in X$ and $\lambda \in [0, 1]$. Affine functions on X are vNM utility functions. Similarly, a function $R : \mathbb{M} \to \mathbb{R}$ is affine if $R(\lambda A + (1 - \lambda)B) = \lambda R(A) + (1 - \lambda)R(B)$ for all $A, B \in \mathbb{M}$ and $\lambda \in [0, 1]$.

Definition 1. A *preference over choices (PC) representation* of $\succsim$ is a triple (u, v, r) such that $\succsim$ is represented by $U(x, A) = u(x) + v(x) - v(r(A))$ where

(i) u and v are affine functions on X;
(ii) $r : \mathbb{M} \to X$ satisfies $r(A) \in \text{conv}(A)$ for every $A \in \mathbb{M}$, and $A \mapsto v(r(A))$ is continuous and affine.

Since $r(\{x\}) = x$, the term $u(x) = U(x, \{x\})$ is the value of x when it is imposed: u represents the evaluator's ranking of outcomes, the consequentialist's part. The judge's part,

$$Q(x, A) \coloneqq v(x) - v(r(A)),$$

is called the *quality* of (x, A), which compares the chosen option with the *benchmark* $r(A)$ according to the *ideal ranking* v. I call $R(A) \coloneqq v(r(A))$ the *benchmark value* of A. The quality of an imposed choice is always zero: $Q(x, \{x\}) = 0$ for all x. Hence, I say (x, A) earns praise if $Q(x, A) > 0$ and blame if $Q(x, A) < 0$.[15]

[15] The functions u and v need not be aligned. In a variety of contexts, the evaluator may prefer outcome x to outcome y, while regarding the choice $(x, \{x, y\})$ as blameworthy.

Theorem 1. $\succsim$ *satisfies Axioms 1–4 if and only if it has a PC representation* (u, v, r). *Moreover,* (u', v', r') *is another PC representation of* $\succsim$ *if and only if* $u' = \alpha u + \beta_1$ *and* $v' = \alpha v + \beta_2$ *for some* $\alpha > 0$ *and* $\beta_1, \beta_2 \in \mathbb{R}$, *and* $v(r'(A)) = v(r(A))$ *for every* $A \in \mathbb{M}$.

The ranking of outcomes and the ideal ranking are thus identified up to a common positive affine transformation. The benchmark of each menu is identified up to its benchmark value: any lottery in $\text{conv}(A)$ with the same value under v can serve as the benchmark of A. Section 5 derives the representation from the axioms, with the remaining details in Appendix A.

2.4. Representation 2: The Preference over Difficult Choices

The PC representation leaves the benchmark value, $v(r(A))$, largely unrestricted, apart from boundedness and affinity. The following axiom gives it further behavioral content, yielding an explicit form that captures the evaluator's preference for difficult choices over easy choices.

Axiom 5 (Choice Betweenness)**.** $(x, A) \succsim (x, B)$ *implies* $(x, A) \succsim (x, A \cup B) \succsim (x, B)$.

When the same option is chosen from two menus and from their union, Choice Betweenness (CB) requires the choice from the union to be judged between the other two. In terms of difficulty, if choosing from A is harder than choosing from B, adding the options of B to A cannot make the choice harder, and adding the options of A to B cannot make it easier.

Definition 2. A *preference over difficult choices (PDC) representation* of $\succsim$ is a triple (u, v, τ) of affine functions on X such that $\succsim$ is represented by

$$U(x, A) = u(x) + v(x) - \Big[\max_{y \in A} \big(v(y) + \tau(y)\big) - \max_{z \in A} \tau(z)\Big].$$

For each menu A, let $\hat{x}(A)$ be a maximizer of $v + \tau$ over A, and let $c(A) \coloneqq \max_{z \in A} \tau(z) - \tau(\hat{x}(A)) \geq 0$. The judge's part of a PDC representation then takes the form

$$Q(x, A) = v(x) - v(\hat{x}(A)) + c(A).$$

As in the introduction, I interpret τ as the evaluator's belief about how much the DM values each alternative, $\hat{x}(A)$ as the choice she expects him to make, and $c(A)$ as the cost she expects that choice to impose on him. Hence, praise and blame have two sources: surprise, $v(x) - v(\hat{x}(A))$, and expected cost, $c(A)$.[16] Since the PDC is a special case of the

[16] If $v + \tau$ has several maximizers in A, surprise and expected cost depend on which one $\hat{x}(A)$ denotes. This does not change the judgment itself and thus leaves every conclusion about praise and blame intact.

PC representation, the benchmark value of A is $R(A) = v(\hat{x}(A)) - c(A)$: the value of the expected choice, minus its expected cost.[17]

Theorem 2. *$\succsim$ satisfies Axioms 1–5 if and only if it has a PDC representation (u, v, τ). Moreover, if (u, v, τ) is a PDC representation of $\succsim$, the following are equivalent:*

(i) every PDC representation of $\succsim$ equals $(\alpha u + \beta_1,\ \alpha v + \beta_2,\ \alpha\tau + \beta_3)$ for some $\alpha > 0$ and $\beta_1, \beta_2, \beta_3 \in \mathbb{R}$;

(ii) $(x, A) \succ (x, \{x\})$ and $(y, \{y\}) \succ (y, B)$ for some $(x, A), (y, B) \in \mathbb{C}$.

Condition (ii) says that the evaluator praises some choice and blames some choice. It involves only $\succsim$ and can therefore be checked without knowing the representation. When it fails, different beliefs τ yield the same preference (see Section 3.2). Section 5.3 derives the representation, building on Gul and Pesendorfer (2001, Theorem 1), and Appendix B proves the uniqueness part, building on their Theorem 4.[18]

3. Praise, Blame, and Difficulty

This section studies how the PDC representation assigns praise and blame. Section 3.1 derives the judgment of each choice from a menu with two options, which formalizes the defendant example of the introduction. Section 3.2 identifies the evaluators whose benchmark follows a fixed rule, such as the midpoint used in theories of reciprocity. Section 3.3 shows how the PDC representation both subsumes and departs from fixed rules.

3.1. Two-Option Menus

Consider a menu $\{x, y\}$ from which the evaluator wants the DM to choose x, that is, $v(x) > v(y)$. Let $s := v(x) - v(y)$ denote what is at stake in the choice according to her ideal ranking, and let $k := \tau(y) - \tau(x)$ denote the cost she believes choosing x imposes on him. Then $k < 0$ means that choosing x also serves the DM's own interest.

Proposition 1. *Let (u, v, τ) be a PDC representation, and $v(x) > v(y)$. Then*

$$Q(x, \{x, y\}) = \min\{\max\{k, 0\},\ s\} \qquad \text{and} \qquad Q(y, \{x, y\}) = Q(x, \{x, y\}) - s.$$

[17] If $\hat{x}$ and z^* maximize $v + \tau$ and τ over A, respectively, then $R(A) \leq v(\hat{x}) \leq \max_{x \in A} v(x)$ and $R(A) \geq v(z^*) + \tau(z^*) - \tau(z^*) \geq \min_{x \in A} v(x)$. The benchmark value thus lies between the least and the most ideal values in A.

[18] Under a weaker continuity axiom, Gul and Pesendorfer (2001, Theorem 3) also characterize what they call temptation without self-control. Its counterpart here is the benchmark value $R(A) = \max\{v(x) : x \in A,\ \tau(x) \geq \tau(y) \text{ for all } y \in A\}$: the evaluator always expects the DM to choose an option he values most, and she never expects a choice to be costly for him. Continuity rules out this case.

Proof. See Appendix B. *Q.E.D.*

The choice $(x, \{x, y\})$ earns praise equal to its cost, truncated below at zero and above at the stakes, and choosing y is judged lower by the stakes. Three cases arise.

- If $k \leq 0$, then $\hat{x}(\{x, y\}) = x$ and $c(\{x, y\}) = 0$. Choosing x costs the DM nothing. The evaluator expects x, which earns no praise, and choosing y is blamed by the full stakes, s. That is, $Q(x, \{x, y\}) = 0$ and $Q(y, \{x, y\}) = -s$.

- If $0 < k < s$, then $\hat{x}(\{x, y\}) = x$ and $c(\{x, y\}) = k$. She still expects x but believes it costs him k. Choosing x earns praise k, and choosing y is blamed by $s - k$. That is, $Q(x, \{x, y\}) = k$ and $Q(y, \{x, y\}) = -(s - k)$.

- If $k \geq s$, then $\hat{x}(\{x, y\}) = y$ and $c(\{x, y\}) = 0$. The cost is at least as large as the stakes, and she expects y. Choosing y is judged exactly as if it had been imposed, and choosing x earns praise s. That is, $Q(x, \{x, y\}) = s$ and $Q(y, \{x, y\}) = 0$.

Let x denote resisting the threat and y denote causing the harm, with $(y, \{y\})$ interpreted as causing the harm accidentally. Then the above are the three cases of the defendant in the introduction: refusing costs him nothing, refusing is costly but tolerable, and the threat is of death. The proposition also covers the case in which the defendant chooses the lesser of two evils, with x causing a harm and y a greater harm. Causing the lesser harm earns praise only if it costs him something; when it also saves his own skin ($k \leq 0$), it earns no more praise than the accident.

Finally, the evaluator blames $(y, \{x, y\})$ if and only if she believes that $k < s$, and praises $(x, \{x, y\})$ if and only if she believes that $k > 0$. As in the introduction, two evaluators with the same ideal ranking therefore disagree about blame exactly when only one believes that the cost is below the stakes. They disagree about praise exactly when only one believes that x is costly.

3.2. Fixed Benchmarks

Rabin (1993) and Dufwenberg and Kirchsteiger (2004) measure kindness against a benchmark fixed in advance: in its simplest form, the midpoint between the most and the least that the other player could have received (Section 4 gives their exact measures). More generally, suppose the benchmark of a PC representation is a *fixed fraction* $\theta \in [0, 1]$, that is,

$$R(A) = \theta \max_{x \in A} v(x) + (1 - \theta) \min_{x \in A} v(x) \qquad \text{for every } A \in \mathbb{M}.$$

On every menu, the benchmark then lies the same fraction θ of the way from the least ideal to the most ideal option. The next result identifies the PDC representations with this property.

Proposition 2. *Let (u, v, τ) be a PDC representation with nonconstant v. Its benchmark is a fixed fraction if and only if $\tau = av + \delta$ for some $a, \delta \in \mathbb{R}$. In that case, $\theta = 1$ if $a \geq 0$, $\theta = 1 + a$ if $-1 < a < 0$, and $\theta = 0$ if $a \leq -1$.*

Proof. See Appendix B. *Q.E.D.*

A fixed benchmark thus corresponds to a belief τ that is a rescaled copy of the evaluator's own ideal ranking: beyond that ranking, τ contributes a single number, a, which in turn determines the benchmark.

- If $a \geq 0$, the evaluator believes the DM values the options in the same order as she does, or is indifferent among them. Choosing the most ideal option never costs him anything, and she expects him to choose it. The benchmark is the most ideal option, and no choice earns praise.

- If $-1 < a < 0$, she believes he values the options in the reverse order of her ideal ranking, at $|a|$ times its intensity. She still expects the most ideal option but believes it costs him $|a|$ times the gap between the most and the least ideal options: $c(A) = |a|(\max_{x \in A} v(x) - \min_{x \in A} v(x))$. As a approaches -1, the benchmark approaches the least ideal option. At $a = -\frac{1}{2}$, the benchmark is the midpoint: $\theta = \frac{1}{2}$.

- If $a \leq -1$, the reverse ranking is at least as intense as the ideal ranking. She expects the least ideal option, which costs him nothing. The benchmark is the least ideal option, and no choice is blamed.

Proposition 2 also explains condition (ii) of Theorem 2. The evaluators in the first and third cases ($a \geq 0$ and $a \leq -1$) never praise and never blame, respectively, and within each of these ranges, different values of a yield the same preference. In the second case ($-1 < a < 0$), which includes the midpoint, the evaluator both praises and blames, in which case Theorem 2 identifies τ up to the transformation in its part (i).

The lack of identification at the two ends reflects Relativity. Because the benchmark lies between the least and the most ideal options, no choice is praised or blamed by more than the gap between them; larger praise or blame would require a benchmark outside this range, which Relativity rules out. Once the benchmark reaches the most ideal option ($a \geq 0$) or the least ideal one ($a \leq -1$), a more intense belief τ cannot move it further, and its intensity, a, leaves no trace in the evaluator's judgments.

3.3. Beyond Fixed Benchmarks

In every PDC representation, an added option changes the judgment of other choices only through the expected choice and its expected cost. Under a fixed benchmark, these depend on a menu only through its most and least ideal options, and an added option therefore matters only through its value under v. The next result compares fixed benchmarks with PDC representations in general.

Proposition 3. *Let $B \in \mathbb{M}$, and let $x^*, x_* \in X$ satisfy $v(x^*) \geq v(y) > v(x_*)$ for every $y \in B$.*

(a) If the benchmark of a PC representation is a fixed fraction θ, then, for every $y \in B$,

$$Q(y, B \cup \{x^*\}) \leq Q(y, B) \leq Q(y, B \cup \{x_*\}),$$

where the first inequality is strict if $\theta > 0$ and $v(x^*) > v(y)$ for every $y \in B$, and the second is strict if $\theta < 1$. Moreover, $Q(x^*, B \cup \{x^*\}) \geq Q(y, B)$ for every $y \in B$.

(b) In a PDC representation, if $\tau(x^*) \geq \tau(y)$ for every $y \in B$, then $Q(y, B \cup \{x^*\}) \leq 0$ for every $y \in B \cup \{x^*\}$, with equality at $y = x^*$. If $w \in X$ satisfies $(v + \tau)(w) \leq \max_{y \in B}(v + \tau)(y)$ and $\tau(w) \leq \max_{y \in B} \tau(y)$, then $Q(y, B \cup \{w\}) = Q(y, B)$ for every $y \in B$.

Proof. See Appendix B. *Q.E.D.*

Under a fixed benchmark, part (a) shows, adding a more ideal option (x^*) raises the benchmark: every choice from B is judged no better than before, while choosing the new option is judged at least as well as any choice from B was. Adding a less ideal option (x_*) lowers the benchmark, and every choice from B is judged at least as well as before. Under the PDC representation, part (b) shows, the effect depends on what the DM is believed to value. If he is believed to value the more ideal option at least as much as every option in B, choosing it is easy: the evaluator expects it, it costs him nothing, and no choice earns praise. Adding an option w that affects neither the expected choice nor its expected cost changes no judgment. For instance, if $\tau(x_*) \leq \tau(z)$ for some $z \in B$, then $Q(y, B \cup \{x_*\}) = Q(y, B)$ for every $y \in B$.

The two parts differ because an intermediate fixed fraction, $0 < \theta < 1$, restricts the belief to $\tau = av + \delta$ with $-1 < a < 0$ (Proposition 2). A more ideal option is then always one the DM values less, and a less ideal option is always one he values more: $\tau(x^*) \leq \tau(y) < \tau(x_*)$ for every $y \in B$. Part (b) covers the beliefs that this restriction rules out.

Return to the defendant of Section 3.1, who can refuse the threat, x, or give in and cause the harm, y. Fix the judge's ideal ranking and her assessment of this menu: $v(x) > v(y)$, and refusing is costly but tolerable, $0 < \tau(y) - \tau(x) < v(x) - v(y)$. Now add one of two options: a safe way out, x^*, which avoids the harm and spares him the risk, or a fight against the threat, x_*, after which the harm occurs anyway and he is injured as well. The ideal ranking orders them $v(x^*) > v(x) > v(y) > v(x_*)$, and only the judge's belief about the added options remains to be specified. In the PDC representation, she can believe that he values the safe way out most and the fight least, $\tau(x^*) > \tau(y) > \tau(x) > \tau(x_*)$. By part (b), escaping is then morally neutral, and refusing at personal risk no longer earns praise. By part (b) again, the fight changes neither the choice she expects nor its cost, and she assesses the choices of x and y exactly as before.

A fixed fraction leaves no such freedom. By Proposition 2, it requires $\tau = av + \delta$, and her assessment of the original menu then fixes $a = -[\tau(y) - \tau(x)]/[v(x) - v(y)]$, which lies between -1 and 0 because the threat is costly but tolerable. The judge must then believe that the defendant values every option in the reverse order of her ideal ranking, $\tau(x^*) < \tau(x) < \tau(y) < \tau(x_*)$: he wants the safe way out least and the futile fight most. For every such a, part (a) implies that she praises escaping more than she praises refusing when the escape is not available. It also implies that when the fight is available, she judges both refusing and giving in more favorably than when it is not, because she believes he wants the fight most. Causing the harm even earns praise if $(1 + a)[v(x) - v(y)] < -a[v(y) - v(x_*)]$.

The belief in the PDC representation above ranks the safe way out and the fight in the same order as the ideal ranking: the safe way out is best by both, and the fight is worst by both. A fixed fraction consistent with the original menu cannot hold such a belief, because it must rank every option in the reverse order of the ideal ranking. Judging choices from menus with such options requires a belief that is not tied to v.

4. Measures of Kindness in Theories of Reciprocity

Theories of intention-based reciprocity specify, through a kindness term, how a player judges another player's choice. This section compares these terms with the two representations. Let each sure alternative be an allocation of money between a chooser, who plays the role of the DM, and a recipient, who plays the role of the evaluator. For each $x \in X$, write $p_c(x)$ and $p_r(x)$ for their expected payoffs, and call x an allocation. In these theories, the allocations available to the chooser depend on the players' beliefs about each other's strategies; I hold these beliefs fixed, so that he chooses x from a fixed menu A. Write $\max_A p_r$ and $\min_A p_r$ for the most and the least the recipient could receive, and $\max_A p_c$ for the most the chooser

could obtain. The recipient's payoff serves as the ideal ranking: $v = p_r$.

For concreteness, when I apply the PDC representation, I take her belief to be $\tau = bp_c$ for some $b \geq 0$: she believes that the chooser values only his own payoff, with weight b relative to her ideal ranking. Then $Q(x, A) = p_r(x) - R(A)$, where

$$R(A) = p_r(\hat{x}(A)) - b\Big[\max_A p_c - p_c(\hat{x}(A))\Big]$$

and $\hat{x}(A)$ maximizes $p_r + bp_c$ over A. She expects the chooser to maximize this weighted sum of payoffs, at a cost to him of b times the payoff he gives up relative to the most he could obtain.

Outcome and judgment. In the PC representation, a choice is judged relative to having had its outcome imposed, and any attitude toward the allocation itself belongs to the ranking of outcomes, $U(x, \{x\})$. Falk and Fischbacher (2006) do not make this separation. They measure kindness by the payoff difference,

$$K_{\text{FF}}(x, A) = \eta(x, A)\big[p_r(x) - p_c(x)\big],$$

where the intention factor $\eta(x, A)$ equals 1 if the chooser could have given the recipient less, when $p_r(x) \geq p_c(x)$, or more, when $p_r(x) < p_c(x)$, and equals a constant $\varepsilon \in [0, 1]$ otherwise. Since $\eta(x, \{x\}) = \varepsilon$, an allocation the chooser could not avoid is judged kind or unkind when $\varepsilon > 0$, and relative to having had no choice, $K_{\text{FF}}(x, A) - K_{\text{FF}}(x, \{x\}) = [\eta(x, A) - \varepsilon][p_r(x) - p_c(x)]$. The forgone allocations thus affect only the magnitude of kindness, through η, while the payoff difference sets its sign: a choice can be kind only if the recipient receives more than the chooser, and unkind only if she receives less. Let $A = \{x, y, z\}$, where x destroys \$10 of the recipient's money, y destroys \$10 of the chooser's own money, and z gives each of them \$100. Their term judges y the kindest choice because only y gives the recipient more than the chooser: $K_{\text{FF}}(y, A) = 10 > K_{\text{FF}}(z, A) = 0 > K_{\text{FF}}(x, A) = -10$. In the PDC representation, whatever b is, the recipient expects z, which costs the chooser nothing. Choosing z earns no praise, and the other two choices are blamed for falling short of it: $Q(z, A) = 0 > Q(y, A) = -100 > Q(x, A) = -110$. A concern with who gets more than whom, such as envy, can still enter the recipient's ranking of outcomes, $U(x, \{x\})$, but it no longer determines whether a choice is praised or blamed.

The midpoint. Rabin (1993) and Dufwenberg and Kirchsteiger (2004) measure kindness against the midpoint between the most the recipient could receive and the least she could

receive at an efficient allocation:

$$K_{\mathrm{DK}}(x, A) = p_r(x) - \tfrac{1}{2}\Big[\max_A p_r + \min_{E(A)} p_r\Big],$$

where $E(A)$ is the set of efficient allocations in A. An allocation in A is efficient if no other allocation in A gives one player more and the other no less, and it is dominated otherwise.[19] Apart from the efficiency restriction, to which I return below, the midpoint is the fixed fraction $\theta = \frac{1}{2}$ of Section 3.2. By Proposition 2, it is therefore the benchmark of a PDC representation if and only if $\tau = -\frac{1}{2}p_r + \delta$ for some constant δ: the recipient believes that the less an allocation gives her, the more the chooser values it. The midpoint looks reasonable in constant-sum games, such as a dictator game, because in such a game this belief is equivalent to $\tau = \frac{1}{2}p_c$.[20] Because the midpoint benchmark requires the evaluator to believe that the DM ranks every option in the reverse order of her ideal ranking, it is less reasonable outside constant-sum games, and beyond allocation games. Dufwenberg and Kirchsteiger (2004, footnote 7) choose the midpoint for its simplicity and performance; Proposition 2 shows what it assumes about the chooser.

Efficient allocations. The efficiency restriction in K_{DK} takes the minimum over the efficient allocations, $\min_{E(A)} p_r$, rather than over all of A; a dominated allocation therefore cannot lower the midpoint. In a PC representation whose benchmark is a fixed fraction $\theta < 1$, adding a dominated allocation can lower the benchmark (Proposition 3(a)). In the PDC representation with $\tau = bp_c$, by contrast, the efficiency restriction is implied rather than imposed: an added allocation that changes neither the expected choice nor its expected cost changes no judgment of the other choices (Proposition 3(b)), and a dominated allocation changes neither. The efficiency restriction in K_{DK}, however, excludes only dominated allocations. An allocation that gives the recipient more than any other raises the midpoint even when she would not expect the chooser to choose it, whereas under the belief $\tau = bp_c$, such an allocation changes neither the expected choice nor its expected cost and changes no judgment.

[19] The term of Rabin (1993) is K_{DK} divided by $\max_A p_r - \min_A p_r$. The division changes the magnitude of kindness but not its sign.

[20] Indeed, $p_r = S - p_c$ for a constant S, and thus $\frac{1}{2}p_c = -\frac{1}{2}p_r + \frac{1}{2}S$. Since $\frac{1}{2}p_c$ has the form $-\frac{1}{2}p_r + \delta$ and no allocation of a constant-sum game is dominated, K_{DK} is exactly the judge's part of the PDC representation with $\tau = \frac{1}{2}p_c$. More generally, if $p_c = S - \gamma p_r$ for constants S and $\gamma > 0$, the belief $\tau = \frac{1}{2\gamma}p_c$ yields the midpoint in the same way. Such payoffs form a linear Pareto frontier, the case in which Rabin (1993, p. 1286) motivates the midpoint as the payoff the recipient receives when the chooser "splits the difference" with her.

Sacrifice. Jang and Sohn (2024) multiply K_{DK} by a factor that increases with the chooser's sacrifice, the payoff he gives up relative to the most he could obtain:

$$K_{\text{JS}}(x, A) = K_{\text{DK}}(x, A)\Big[1 + \max_{A} p_c - p_c(x)\Big].$$

Since this factor is at least one, sacrifice increases the magnitude of kindness without changing its sign, which the midpoint still sets. Moreover, because sacrifice is measured at the allocation chosen, the cost of choosing kindly raises the kindness of that choice but does not lessen the unkindness of choosing otherwise.[21] For example, let $A = \{y_1, y_2\}$ with $y_1 = (10, 0)$ and $y_2 = (10 - \kappa, 10)$, where the first entry is the chooser's payoff and $0 < \kappa < 10$. Since both allocations are efficient, the midpoint is \$5, and

$$K_{\text{JS}}(y_2, A) = 5(1 + \kappa) \qquad \text{and} \qquad K_{\text{JS}}(y_1, A) = -5.$$

In the PDC representation, by contrast, the expected cost is separate from surprise. If $b\kappa < 10$, the recipient expects y_2, the expected cost is $c(A) = b\kappa$, and

$$Q(x, A) = \big[p_r(x) - p_r(y_2)\big] + b\kappa \qquad \text{for } x \in A,$$

where the first term is surprise and the second is expected cost. Hence $Q(y_2, A) = b\kappa$ and $Q(y_1, A) = b\kappa - 10$ (Proposition 1): for $b > 0$, a higher κ raises the quality of both choices, as the principle of Nelkin (2016) requires.

Each of these terms fixes in advance the rule that sets the sign of kindness: the midpoint of the recipient's payoffs, or which player receives more. The intention factor of Falk and Fischbacher (2006) and the sacrifice in Jang and Sohn (2024) change only its magnitude. In the PC representation, whether a choice earns praise or blame is decided by the benchmark, which is derived from the evaluator's preference rather than fixed in advance. In the PDC representation, the benchmark is set by the choice she expects and the cost she expects it to impose.

[21] For the same reason, the term also rewards waste. Consider a menu of three allocations, listing the chooser's payoff first: $(10, 0)$, $(5, 5)$, and $(3, 5)$. The midpoint is \$2.50, and K_{DK} equals 2.5 for both $(5, 5)$ and $(3, 5)$. The factor is 6 for $(5, 5)$ but 8 for $(3, 5)$, and the term judges choosing $(3, 5)$ kinder, 20 against 15, although it gives the chooser \$2 less and the recipient nothing more. In the PC representation with $v = p_r$, the two choices have the same quality, since $Q(x, A) = p_r(x) - R(A)$.

5. From Axioms to Representations

This section explains how the axioms deliver Theorems 1 and 2. In Section 5.1, the expected utility axioms alone split the evaluator's ranking of choices into her ranking of outcomes and a ranking of choices with outcomes held fixed. In Section 5.2, Relativity places the benchmark among the mixtures of the options on the menu and is shown to be equivalent to impartiality toward menus. In Section 5.3, Choice Betweenness turns the benchmark value into the representation of Gul and Pesendorfer (2001).

5.1. Separating the Consequentialist from the Judge

Two implications of the expected utility axioms, Axioms 1–3, underlie Theorem 1. The first is about randomization. The DM can always condition his choice on a random event, such as a coin flip, even if his menu offers no lottery between its options. Recall that one of the conventional implications of the EU theory in a standard setting is that randomization itself has no value. Likewise, if the evaluator's preference over choices conforms to the EU principles, then she finds no value in *giving up* randomization. The following result formalizes this implication.

Lemma 1 (Irrelevance of Forgone Randomization)**.** *Suppose* $\succsim$ *satisfies Axioms 1–3. For each* $A, B \in \mathbb{M}$*,* $B \subseteq \text{conv}(A)$ *implies* $(x, A) \sim (x, A \cup B)$*.*

Proof. See Appendix A. *Q.E.D.*

In particular, $(x, A) \sim (x, \text{conv}(A))$: adding to a menu the mixtures of its options changes no judgment. To see why, let $A = \{x, y\}$. Mixing (x, A) with $(x, \text{conv}(A))$, each with probability one half, yields $(x, \text{conv}(A))$ itself, because $\frac{1}{2}A + \frac{1}{2}\text{conv}(A) = \text{conv}(A)$. Mixing $(x, \text{conv}(A))$ with itself also yields $(x, \text{conv}(A))$. Independence therefore rules out a strict ranking between (x, A) and $(x, \text{conv}(A))$. The lemma applies to any randomization the DM forgoes, not only to the full set $\text{conv}(A)$. A DM who can condition his choice from A on a fair coin, for example, faces the menu $A_2 := \frac{1}{2}A + \frac{1}{2}A$. Since $A \subseteq A_2 \subseteq \text{conv}(A)$, Lemma 1 gives $(x, A) \sim (x, A_2)$: choosing x without using the coin is judged as choosing x when no coin is available. More generally, the DM may condition his choice on any external state, such as the weather or his mood, that leaves his menu unchanged; under the EU axioms, such conditioning is irrelevant to the evaluator's judgment.[22]

[22] In contrast, she pays attention to states that alter feasibility. If an illness removes x but not y from his menu with probability one half, choosing x when healthy is the mixture $\frac{1}{2}(x, \{x, y\}) + \frac{1}{2}(y, \{y\})$, which she evaluates by the menu in each state.

The second implication is an affine representation. A function $U : \mathbb{C} \to \mathbb{R}$ is affine if $U(\lambda(x, A) + (1 - \lambda)(y, B)) = \lambda U(x, A) + (1 - \lambda)U(y, B)$ for all $(x, A), (y, B) \in \mathbb{C}$ and $\lambda \in [0, 1]$. The mixture space theorem (Herstein and Milnor, 1953) does not apply to $\mathbb{C}$ directly, because mixing a menu with itself need not return the same menu: $\frac{1}{2}A + \frac{1}{2}A \neq A$ when A is not convex. Lemma 1 addresses this issue.

Proposition 4. $\succsim$ *satisfies Axioms 1–3 if and only if it is represented by a continuous affine function* $U : \mathbb{C} \to \mathbb{R}$.

Proof. See Appendix A. *Q.E.D.*

Choices from convex menus form a mixture space, on which the mixture space theorem yields an affine representation. By Lemma 1, evaluating each choice (x, A) at $(x, \text{conv}(A))$ extends it to all choices.

The affine representation separates the two parts of the evaluator. Take $x, y \in A \cap B$. The mixtures $\frac{1}{2}(x, A)+\frac{1}{2}(y, B)$ and $\frac{1}{2}(y, A)+\frac{1}{2}(x, B)$ are the same choice, $(\frac{1}{2}x+\frac{1}{2}y,\ \frac{1}{2}A+\frac{1}{2}B)$, and their values under U are equal. Rearranging gives

$$U(x, A) - U(y, A) = U(x, B) - U(y, B). \tag{1}$$

How much more the evaluator values choosing x than choosing y from the same menu does not depend on the menu, and how much more she values choosing x from A than from B does not depend on x.

Proposition 5 (Separability)**.** *Let* U *be a continuous affine representation of* $\succsim$*. Then*

$$U(x, A) = u(x) + v(x) - R(A) \qquad \textit{for every } (x, A) \in \mathbb{C},$$

where $u(x) \coloneqq U(x, \{x\})$*,* $R : \mathbb{M} \to \mathbb{R}$ *is continuous and affine, and* $v(x) \coloneqq R(\{x\})$*.*

Proof. Let X be the menu of all lotteries. For a menu A and an option $x \in A$, let $R(A) \coloneqq U(x, X) - U(x, A)$. This does not depend on which $x \in A$ is used: for $x, y \in A$, both options also belong to X, and (1) with $B = X$ gives $U(x, A) - U(y, A) = U(x, X) - U(y, X)$, that is, $U(x, X) - U(x, A) = U(y, X) - U(y, A)$. Now let $u(x) \coloneqq U(x, \{x\})$ and $v(x) \coloneqq R(\{x\})$. Applying the definition of R to the menu $\{x\}$ gives $v(x) = U(x, X) - U(x, \{x\})$, that is, $U(x, X) = u(x) + v(x)$. Hence, for every choice (x, A), $U(x, A) = U(x, X) - R(A) = u(x) + v(x) - R(A)$. The affinity and continuity of the function R follow from the affinity and continuity of U. *Q.E.D.*

Proposition 5 already has the form of the PC representation. The consequentialist's part is $u(x) = U(x, \{x\})$, the value of x when it is imposed, and u represents the evaluator's ranking of outcomes. The judge's part, $Q(x, A) \coloneqq v(x) - R(A)$, is zero when x is imposed. What the expected utility axioms leave open is where $R(A)$ lies relative to the options in A, which Relativity settles in Section 5.2.

The evaluator's preference also isolates the judge's part directly, through the comparison discussed in the introduction.

Definition 3. The *judge's ranking* is the binary relation $\succsim^Q$ on $\mathbb{C}$ defined by $(x, A) \succsim^Q (y, B)$ if $(\frac{1}{2}x + \frac{1}{2}y,\ \frac{1}{2}A + \frac{1}{2}\{y\}) \succsim (\frac{1}{2}y + \frac{1}{2}x,\ \frac{1}{2}B + \frac{1}{2}\{x\})$.

As in the introduction, the two choices yield the same lottery over outcomes and differ only in which option was chosen, and from which menu. The judge's ranking compares how outcomes were chosen, with the outcomes themselves held fixed.

Proposition 6. *Suppose $\succsim$ satisfies Axioms 1–3, and write its representation as in Proposition 5. Then $(x, A) \succsim^Q (y, B)$ if and only if $Q(x, A) \geq Q(y, B)$. In particular, $(x, \{x\}) \sim^Q (y, \{y\})$ for all $x, y \in X$.*

Proof. The first choice in Definition 3 is $\frac{1}{2}(x, A) + \frac{1}{2}(y, \{y\})$, with value $\frac{1}{2}[u(x) + Q(x, A)] + \frac{1}{2}u(y)$. The second is $\frac{1}{2}(y, B) + \frac{1}{2}(x, \{x\})$, with value $\frac{1}{2}[u(y) + Q(y, B)] + \frac{1}{2}u(x)$. *Q.E.D.*

The judge's ranking identifies the ideal ranking. By Proposition 6, $(x, \{x, y\}) \succsim^Q (y, \{x, y\})$ if and only if $v(x) \geq v(y)$: the evaluator prefers the DM to give up y for x rather than x for y exactly when v ranks x at least as high as y. The EU axioms give the ideal ranking two further properties. First, v is affine, and the ideal ranking is therefore an expected utility ranking of lotteries. An ideal DM, who always makes the best choice from his menu according to the judge's ranking, is an expected utility maximizer. Second, the axioms are not only a technical convenience; they also impose a notion of rationality on the ideal ranking. By Proposition 6, the comparison between choosing x and choosing y is the same on every menu that contains both, which rules out reversals such as $(x, \{x, y\}) \succ^Q (y, \{x, y\})$ and $(y, \{x, y, z\}) \succ^Q (x, \{x, y, z\})$. The first ranking says that the evaluator's ideal DM would choose x over y from $\{x, y\}$; the second says that he would choose y over x once a third option, z, becomes available. Such a reversal violates the weak axiom of revealed preference, a standard requirement of rational choice.

5.2. Relativity and the Benchmark

Relativity locates $R(A)$ relative to the ideal ranking. By Proposition 5, $U(x, A) - U(x, \{x\}) = v(x) - R(A)$ for every choice (x, A). Thus $(x, A) \succsim (x, \{x\})$ if and only if $v(x) \geq R(A)$, and

$(y, \{y\}) \succsim (y, A)$ if and only if $v(y) \leq R(A)$. Under Axioms 1–3, Relativity therefore holds if and only if $\min_{x \in A} v(x) \leq R(A) \leq \max_{x \in A} v(x)$ for every menu A: the value $R(A)$ lies between the least and the most ideal values in A.

The benchmark of Theorem 1 can now be defined from the evaluator's preference.

Definition 4. A *benchmark* of a menu A is a lottery $r(A) \in \text{conv}(A)$ such that $(r(A), A \cup \{r(A)\}) \sim (r(A), \{r(A)\})$.

Choosing a benchmark is thus judged exactly as if it had been imposed. Under Axioms 1–4, every menu has a benchmark, and the benchmarks of A are the lotteries $z \in \text{conv}(A)$ with $v(z) = R(A)$. Indeed, for $z \in \text{conv}(A)$, Lemma 1 gives $(x, A \cup \{z\}) \sim (x, A)$ for every $x \in A$, and hence $R(A \cup \{z\}) = R(A)$; Proposition 5 then gives $(z, A \cup \{z\}) \sim (z, \{z\})$ if and only if $v(z) = R(A)$. Since v is affine and $R(A)$ lies between the least and the most ideal values in A, some $z \in \text{conv}(A)$ satisfies $v(z) = R(A)$.

Proof of Theorem 1. The "if" part is straightforward. For the "only if" part, Proposition 4 yields a continuous affine representation, and Proposition 5 writes it as $U(x, A) = u(x) + v(x) - R(A)$, with u and v affine and R continuous and affine. Choose a benchmark $r(A)$ of every menu A, which exists by Relativity. Then $v(r(A)) = R(A)$, and $A \mapsto v(r(A))$ is continuous and affine. Hence (u, v, r) is a PC representation. The uniqueness part is proved in Appendix A. *Q.E.D.*

Relativity can also be stated through the judge's ranking alone.

Definition 5. $\succsim$ is *partial toward menus* if there are convex menus A and B, each containing more than one option, such that $(x, A) \succ^Q (y, B)$ for all $x \in A$ and $y \in B$.

An evaluator who is partial toward menus ranks every choice from A above every choice from B, whatever the DM chooses: the menu alone decides the judgment.

Proposition 7. *Suppose $\succsim$ satisfies Axioms 1–3. Then $\succsim$ satisfies Relativity if and only if it is not partial toward menus.*

Proof. See Appendix A. *Q.E.D.*

One direction is immediate. Under Relativity, every menu A contains a choice that the judge ranks at least as high as an imposed outcome and a choice that she ranks at most as high: $Q(x, A) \geq 0 \geq Q(y, A)$ for some $x, y \in A$, since $Q(z, \{z\}) = 0$ for every z. For any two menus A and B, some choice from B is then ranked at least as high as some choice from A, and no menu has every choice ranked above every choice from another.

For the other direction, suppose that every choice from some menu B is blamed: $Q(y, B) < 0$ for every $y \in B$. Fix $b \in B$ and $\lambda \in (0, 1)$, and consider the situation in which b is imposed on the DM with probability λ and he chooses from B with probability $1-\lambda$. Choosing y from B in this situation is the choice $\lambda(b, \{b\}) + (1-\lambda)(y, B)$, whose menu is $A := \lambda\{b\} + (1-\lambda)B$. Since Q is affine and $Q(b, \{b\}) = 0$, $Q(\lambda b + (1-\lambda)y, A) = (1-\lambda)Q(y, B)$. Since $Q(\cdot, B)$ is bounded and its maximum over B is negative, choosing λ close enough to one gives $(1-\lambda)Q(y, B) > Q(y', B)$ for all $y, y' \in B$. The judge then ranks every choice from A above every choice from B: $(x, A) \succ^Q (y', B)$ for all $x \in A$ and $y' \in B$. The case in which every choice from B is praised is symmetric.

By Proposition 7, Relativity is exactly the requirement that the menu alone never earns the DM praise or blame. A violation of Relativity is therefore a sign that the evaluator is interested in external factors, beyond the outcomes and the judgment of the DM's choices. For instance, Relativity rules out evaluators who value the DM's freedom of choice (Suzumura and Xu, 2009) or flexibility (Kreps, 1979) for its own sake, and evaluators who would rather impose an outcome than let the DM choose it, for instance to spare him the effort of choosing or to keep control over his outcomes.

5.3. From Choices to Menus

Theorem 1 turns the remaining question into one about the benchmark value R. For $x \in A \cap B$, the theorem gives $(x, A) \succsim (x, B)$ if and only if $R(A) \leq R(B)$: holding the chosen option fixed, the evaluator prefers the choice from the menu with the lower benchmark value. Choice Betweenness therefore requires $R(A \cup B)$ to lie between $R(A)$ and $R(B)$ whenever A and B share an option. This is the Set Betweenness axiom of Gul and Pesendorfer (2001), applied to R and restricted to menus that share an option. The next result shows that, in the space of lotteries, the restriction is without loss.

Lemma 2. *Suppose $\succsim$ satisfies Axioms 1–5, and let R be its benchmark value. For all $A, B \in \mathbb{M}$, if $R(A) \geq R(B)$, then $R(A) \geq R(A \cup B) \geq R(B)$.*

Proof. Let $A' := \frac{1}{2}A + \frac{1}{2}X$ and $B' := \frac{1}{2}B + \frac{1}{2}X$. These menus share an option even when A and B do not: for $x \in A$ and $y \in B$, the lottery $\frac{1}{2}x + \frac{1}{2}y$ belongs to both. Since R is affine, $R(A') = \frac{1}{2}R(A) + \frac{1}{2}R(X)$ and $R(B') = \frac{1}{2}R(B) + \frac{1}{2}R(X)$, and $R(A) \geq R(B)$ implies $R(A') \geq R(B')$. Choice Betweenness then gives $R(A') \geq R(A' \cup B') \geq R(B')$. Since $A' \cup B' = \frac{1}{2}(A \cup B) + \frac{1}{2}X$, affinity again yields $R(A) \geq R(A \cup B) \geq R(B)$. *Q.E.D.*

Proof of Theorem 2. The "if" part is straightforward. The benchmark value of a PDC representation is continuous and affine and lies between the least and the most ideal values in

each menu, which yields Axioms 1–4 by Theorem 1; it also satisfies Set Betweenness, which yields Choice Betweenness. For the "only if" part, let (u, v, r) be a PC representation of $\succsim$ with benchmark value R, and let $\succeq_R$ be the binary relation over menus represented by R. Then $\succeq_R$ is complete and transitive. Its upper and lower contour sets are closed because R is continuous, it satisfies the independence axiom of Gul and Pesendorfer (2001) because R is affine, and it satisfies their Set Betweenness by Lemma 2. Their Theorem 1 yields affine functions u^* and v^* on X such that

$$W(A) \coloneqq \max_{y \in A} \big(u^*(y) + v^*(y)\big) - \max_{z \in A} v^*(z)$$

also represents $\succeq_R$. Both R and W are affine. Moreover, $R(A) = R(\mathrm{conv}(A))$ by Lemma 1, and $W(A) = W(\mathrm{conv}(A))$ because the maximum of an affine function over A equals its maximum over $\mathrm{conv}(A)$. On convex menus, which form a mixture space, two affine representations of the same weak order differ by a positive affine transformation, and these extend to all menus in $\mathbb{M}$. Hence $R = aW + b$ for some $a > 0$ and $b \in \mathbb{R}$. At singletons, this gives $v = au^* + b$, and therefore

$$R(A) = \max_{y \in A} \big(v(y) + \tau(y)\big) - \max_{z \in A} \tau(z), \qquad \text{where } \tau \coloneqq av^*.$$

Hence (u, v, τ) is a PDC representation. The uniqueness part is proved in Appendix B. *Q.E.D.*

6. Discussion

This section distinguishes judgment from inference (Section 6.1), discusses other related literature (Section 6.2), and outlines two extensions (Section 6.3).

6.1. Judgment and Inference

The evaluator in this paper judges the DM's choices; she does not learn about him from them. In models of signaling and social image (Bénabou and Tirole, 2006) and of image-conscious choice (Piermont, 2025), the response to a choice depends on what the choice reveals about the DM. In the PDC representation, by contrast, the evaluator holds a single belief τ for every choice she judges: she revises it neither after observing the menu nor after observing the choice. The model does not explain how she forms τ, for instance from the DM's past behavior; Theorem 2 identifies τ from her judgments.

The EU axioms separate judging a choice from inferring the DM's preference from it.

Consider a choice (x, A), an outcome z, and $\lambda \in (0, 1]$. In the choice $\lambda(x, A) + (1-\lambda)(z, \{z\})$, the DM chooses x from A with probability λ, and z is imposed otherwise. In my model, the quality of this choice is

$$Q\big(\lambda x + (1-\lambda)z,\ \lambda A + (1-\lambda)\{z\}\big) = \lambda Q(x, A).$$

However, whatever his reasons, the DM behaves *as if* he prefers x to every other option in A, for every λ.[23] If her judge's ranking depended only on what she infers from this behavior, such as his character or desires, the two choices would have the same quality. Then $Q(x, A) = \lambda Q(x, A)$ for every λ, and hence $Q(x, A) = 0$ for every choice: she would neither praise nor blame. Relativity excludes a further form of inference. An evaluator who responds only to what the DM's menu reveals about him judges every choice from that menu alike. Unless she neither praises nor blames, she then praises or blames every choice from some menu: the menu alone earns the DM praise or blame, which Relativity rules out (Proposition 7).

6.2. Other Related Literature

This section discusses connections to the literature beyond those covered in the introduction. Economists have long recognized that two choices leading to the same outcome may be evaluated differently depending on the menus from which they were made.[24] Several strands of research explore related ideas, but none take as the primitive an observer's ranking of another person's choices by how they compare with the alternatives he gave up.

The closest precedent for using outcome-menu pairs as primitives in an axiomatic approach is Suzumura and Xu (2009), who study preferences for freedom using sure outcomes (see also Sen, 1988; Pattanaik and Xu, 1990, 1998). To my knowledge, my model is the first to employ the structure of lotteries, which is essential for my results, to study outcome–menu pairs in a similar style. The two frameworks also differ in whose ranking they study and why. Their agent ranks his own opportunities and may prefer (x, A) to (y, B) whenever A contains more alternatives than B. Here, the ranking belongs to an observer judging another person's choices, and by Relativity, the number of opportunities alone never earns the DM praise or blame.[25]

[23] In the terminology of Piermont (2025), the two choices induce the same image.

[24] Regret (Bell, 1982; Loomes and Sugden, 1982) is a classic example. Contextual features (e.g., whether an option is the last one available in the market) can also be influential (see Dietrich and List, 2016; Shafir et al., 1993).

[25] See also Piermont (2025), who studies the DM's image inferred from his choice, and Manzini et al. (2026), who study the opportunity cost of a choice. Both ask how these menu-dependent features determine what the DM chooses, not how his choice is judged.

A separate literature in philosophy and economics studies higher-order preferences: preferences over one's own preferences, interpreted as desires about what to desire (Frankfurt, 1971; Jeffrey, 1974). This approach is largely normative and self-reflective: its objects are preference relations, and its question is what a person should want to want (Nehring, 2006; Hayashi, 2024; Pivato, 2024, 2025). The evaluator in this paper instead ranks actual choices. Her judgments are meant in the accountability sense: she holds the DM responsible for what he did, not for his inner character.

This paper also connects to the literature on menu preferences in a way the introduction does not discuss. Suppose the evaluator and the DM are the same person, so that the DM judges his own choices. His preference over menus can then be derived from that judgment: $M(A) \coloneqq \max_{x \in A} U(x, A)$.[26] Several representations of menu preferences are special cases of M. For example, Gul and Pesendorfer (2001)'s preference for commitment is obtained when $v(r(A)) = \max_{y \in A} v(y)$ for all A. Kopylov (2012)'s representation for a perfectionist who benefits from having normative goals on his menu is another special case. It takes the form

$$M_K(A) \coloneqq \max_{x \in A} [\bar{u}(x) + \bar{v}(x)] - [\max_{y \in A} \bar{v}(y) - \kappa \max_{z \in A} \bar{u}(z)],$$

where $\kappa \geq 0$. Let $U(x, \{x\}) = (1+\kappa)\bar{u}(x)$, $v(x) = \bar{v}(x) - \kappa\bar{u}(x)$, and $v(r(A)) = \max_{y \in A} \bar{v}(y) - \kappa \max_{z \in A} \bar{u}(z)$. Then $M = M_K$. Because these representations differ only in their benchmarks, M places them in a common structure, in which one can ask whether a DM's preference over menus reflects a judgment of his own choices or some other motive. For M_K, when $\kappa < 0$, which his model allows, a menu A can satisfy $M_K(A) < M_K(\{x\})$ for all $x \in A$, which Relativity rules out. In this case, the perfectionist is averse to the conflict he faces when choosing among alternatives that are appealing for different reasons. This aversion may reflect discomfort with having to judge his own choices, rather than reflect the judgment itself. Yet I do not emphasize this self-evaluation interpretation, for the reason given in the introduction.

6.3. Extension

I discuss two extensions. First, one could construct a stochastic choice environment that records what the DM would choose from every binary menu. Because the resulting contingency plan matches "behaving according to" an entire preference ranking over all alternatives, evaluating such plans would connect the present choice-based framework to preferences

[26] Under the usual timing, in which the DM first chooses a menu and later chooses from it, this interpretation assumes that his later self treats the menu as given rather than judging his earlier choice of it, an assumption that choices over menus cannot verify (see Bernheim et al., 2024).

over preference relations. More generally, tailoring the choice environment to reveal a particular behavioral trait could generate preferences over risk attitudes or other classes of preferences. Second, the choice domain could be enlarged to allow the DM to declare indifference among a set of options rather than select a single outcome. This distinction matters because the same outcome may be selected under either strict preference or indifference, while the declaration of indifference may itself be evaluated as open-mindedness or instead as indecisiveness or a lack of commitment. Extending the choice space would allow the model to distinguish among evaluators according to how they value these features.[27]

7. Conclusion

This paper characterizes how an observer judges another person's choices by extending standard techniques of decision theory, from EU theory to models of temptation and self-control. Her preference over outcome-menu pairs separates the value she places on outcomes from her praise or blame for choosing them. It also reveals what she thinks ought to be chosen and what she believes the chooser values, and hence how costly she believes doing the right thing is for him. Theories of reciprocity, for example, specify such judgments in advance; deriving them from preferences shows what those specifications assume.

More broadly, the framework makes the judgment of others' choices an object of decision-theoretic study rather than a modeling assumption. It thereby connects models of conflict within a person with social interactions in which rewards, punishment, and regulation respond to the quality of choices as well as to their outcomes.

Appendix

A. Proofs for Section 5

Proof of Lemma 1. Fix $\alpha \in (0, 1/|Z|]$.

Step 1: $\alpha A + (1-\alpha)\text{conv}(A) = \text{conv}(A)$. For $\subseteq$, let $a \in A$ and $w \in \text{conv}(A)$. Since $A \subseteq \text{conv}(A)$ and $\text{conv}(A)$ is convex, $\alpha a + (1-\alpha)w \in \text{conv}(A)$. For $\supseteq$, let $z \in \text{conv}(A)$. Since X has dimension $|Z|-1$, Carathéodory's theorem (Rockafellar, 1970) implies that there exist $m \leq |Z|$ options $x_1, \ldots, x_m \in A$ and weights $\lambda_1, \ldots, \lambda_m \geq 0$ with $\lambda_1 + \cdots + \lambda_m = 1$ such that $z = \lambda_1 x_1 + \cdots + \lambda_m x_m$. The m weights sum to one, and thus some weight

[27] Some preliminary results for both extensions are available upon request. For the first extension, I provide a representation of preference over complete binary relations and illustrate how the same logic generates a preference over risk attitudes. For the second, I reproduce the baseline model in the extended choice space by characterizing a *no preference for indifference* condition and discuss its limitations in the extended framework.

satisfies $\lambda_j \geq 1/m \geq 1/|Z| \geq \alpha$. Define $w := [(\lambda_j - \alpha)x_j + \sum_{i \neq j} \lambda_i x_i]/(1-\alpha)$. Its weights are nonnegative and sum to $(1-\alpha)/(1-\alpha) = 1$ which means $w \in \text{conv}(A)$. Moreover, $\alpha x_j + (1-\alpha)w = \lambda_1 x_1 + \cdots + \lambda_m x_m = z$. Hence $z \in \alpha A + (1-\alpha)\text{conv}(A)$.

Step 2: $(x, A) \sim (x, \text{conv}(A))$. By Step 1, $\alpha(x, A) + (1-\alpha)(x, \text{conv}(A)) = (x, \text{conv}(A))$. Since $\text{conv}(A)$ is convex, $\alpha(x, \text{conv}(A)) + (1-\alpha)(x, \text{conv}(A)) = (x, \text{conv}(A))$ as well. Hence, if $(x, A) \not\sim (x, \text{conv}(A))$, then Independence is violated.

Step 3. If $B \subseteq \text{conv}(A)$, then $\text{conv}(A \cup B) = \text{conv}(A)$, and Step 2 applied to $A \cup B$ gives $(x, A \cup B) \sim (x, \text{conv}(A)) \sim (x, A)$. *Q.E.D.*

Proof of Proposition 4. The "if" part is standard. For the "only if" part, let $\mathbb{C}^c := \{(x, A) \in \mathbb{C} : A \in \mathbb{M} \text{ is convex}\}$. This set is closed under mixtures, and it is a mixture space. The only property that requires an argument is

$$\lambda\big[\mu(x, A) + (1-\mu)(y, B)\big] + (1-\lambda)(y, B) = \lambda\mu(x, A) + (1-\lambda\mu)(y, B)$$

for $\lambda, \mu \in [0, 1]$, whose outcome parts agree trivially. For the menu parts, take $x' \in A$ and $y_1, y_2 \in B$. If $\lambda\mu < 1$, then $\lambda[\mu x' + (1-\mu)y_1] + (1-\lambda)y_2 = \lambda\mu x' + (1-\lambda\mu)y'$ with $y' := [\lambda(1-\mu)y_1 + (1-\lambda)y_2]/(1-\lambda\mu)$, which belongs to B because B is convex; setting $y_1 = y_2$ gives the reverse inclusion. The case $\lambda\mu = 1$ is immediate.

The restriction of $\succsim$ to $\mathbb{C}^c$ satisfies the axioms of the mixture space theorem (Herstein and Milnor, 1953), which yields an affine function $\hat{U}$ on $\mathbb{C}^c$ that represents it; Continuity makes $\hat{U}$ continuous. Define $U(x, A) := \hat{U}(x, \text{conv}(A))$. By Lemma 1, $(x, A) \sim (x, \text{conv}(A))$, and U therefore represents $\succsim$ on $\mathbb{C}$. Since $\text{conv}(\lambda A + (1-\lambda)B) = \lambda\text{conv}(A) + (1-\lambda)\text{conv}(B)$ and $A \mapsto \text{conv}(A)$ is continuous in the Hausdorff metric, U is affine and continuous. *Q.E.D.*

Proof of the uniqueness part of Theorem 1. The "if" direction follows by substitution. For the "only if" direction, let (u, v, r) and (u', v', r') be PC representations of $\succsim$, with U and U' the corresponding functions. Both are continuous affine representations of $\succsim$. On $\mathbb{C}^c$, which is a mixture space, they differ by a positive affine transformation, and by Lemma 1 each is determined by its values on $\mathbb{C}^c$. Hence $U' = \alpha U + \beta_1$ for some $\alpha > 0$ and $\beta_1 \in \mathbb{R}$. At imposed outcomes, this gives $u' = \alpha u + \beta_1$. For $x, y \in X$, comparing $U'(x, X) - U'(y, X)$ with $\alpha[U(x, X) - U(y, X)]$ gives $v'(x) - v'(y) = \alpha[v(x) - v(y)]$, and hence $v' = \alpha v + \beta_2$ for some $\beta_2 \in \mathbb{R}$. Finally, for $x \in A$,

$$v'(r'(A)) = u'(x) + v'(x) - U'(x, A) = \alpha\big[u(x) + v(x) - U(x, A)\big] + \beta_2 = \alpha v(r(A)) + \beta_2,$$

while $v'(r'(A)) = \alpha v(r'(A)) + \beta_2$. Hence $v(r'(A)) = v(r(A))$. *Q.E.D.*

Proof of Proposition 7. Write the representation of $\succsim$ as in Proposition 5; by Proposition 6, $\succsim^Q$ is represented by $Q(x, A) = v(x) - R(A)$.

Suppose first that $\succsim$ satisfies Relativity, and let A and B be menus. Since $R(A)$ and $R(B)$ then lie between the least and the most ideal values in their menus (Section 5.2), $\min_{x\in A} Q(x, A) = \min_{x\in A} v(x) - R(A) \leq 0 \leq \max_{y\in B} v(y) - R(B) = \max_{y\in B} Q(y, B)$. Hence $(y, B) \succsim^Q (x, A)$ for some $x \in A$ and $y \in B$, and $\succsim$ is not partial toward menus.

Conversely, suppose that $\succsim$ violates Relativity. Then (Section 5.2) some menu B satisfies $R(B) > \max_{x\in B} v(x)$ or $R(B) < \min_{x\in B} v(x)$. Consider the first case; the second is symmetric. By Lemma 1, $R(\mathrm{conv}(B)) = R(B)$, and v attains the same extreme values on B and on $\mathrm{conv}(B)$; replacing B with $\mathrm{conv}(B)$ therefore preserves the violation. Moreover, B contains more than one option because $R(\{x\}) = v(x)$. Assume, then, that B is convex. Let $\varepsilon := R(B) - \max_{x\in B} v(x) > 0$ and $M := R(B) - \min_{x\in B} v(x)$. Fix $b \in B$ and $\lambda \in (0, 1)$ with $(1-\lambda)M < \varepsilon$, and let $A := \lambda\{b\} + (1-\lambda)B$, a convex menu with more than one option. Each option of A has the form $\lambda b + (1 - \lambda)y$ with $y \in B$, and since $R(A) = \lambda v(b) + (1 - \lambda)R(B)$,

$$Q\big(\lambda b + (1 - \lambda)y, A\big) = (1 - \lambda)\big[v(y) - R(B)\big] \geq -(1 - \lambda)M > -\varepsilon \geq Q(y', B)$$

for every $y' \in B$. Every choice from A is therefore ranked above every choice from B by $\succsim^Q$, and $\succsim$ is partial toward menus. *Q.E.D.*

B. Proofs for Sections 2.4 and 3

Throughout, (u, v, τ) is a PDC representation of $\succsim$, and

$$R(A) = \max_{y\in A} \big(v(y) + \tau(y)\big) - \max_{z\in A} \tau(z)$$

is its benchmark value.

Lemma 3. *Let f and g be affine functions on X such that $f(x) \neq f(y)$ for some $x, y \in X$. If $g(x) \geq g(y)$ for all $x, y \in X$ with $f(x) > f(y)$, then $g = af + b$ for some $a \geq 0$ and $b \in \mathbb{R}$.*

Proof. Choose $\underline{z}, \overline{z} \in X$ minimizing and maximizing f, respectively. Since f is nonconstant, $f(\underline{z}) < f(\overline{z})$. First, let $x, y \in X$ satisfy $f(x) = f(y)$. For every $\varepsilon \in (0, 1)$, the affinity of f gives $f((1 - \varepsilon)x + \varepsilon\overline{z}) > f((1 - \varepsilon)y + \varepsilon\underline{z})$. Hence $g((1 - \varepsilon)x + \varepsilon\overline{z}) \geq g((1 - \varepsilon)y + \varepsilon\underline{z})$, and the affinity of g gives $(1 - \varepsilon)g(x) + \varepsilon g(\overline{z}) \geq (1 - \varepsilon)g(y) + \varepsilon g(\underline{z})$. Taking limits as $\varepsilon \to 0$ gives $g(x) \geq g(y)$, and interchanging x and y gives the reverse inequality. Thus, $f(x) = f(y)$ implies $g(x) = g(y)$. Now take any $x \in X$ and set $t = [f(x) - f(\underline{z})]/[f(\overline{z}) - f(\underline{z})] \in [0, 1]$. By affinity, $f(x) = f((1 - t)\underline{z} + t\overline{z})$, which implies $g(x) = g((1 - t)\underline{z} + t\overline{z})$. The affinity of g

then gives $g(x) = (1-t)g(\underline{z}) + tg(\overline{z}) = af(x) + b$, where $a = [g(\overline{z}) - g(\underline{z})]/[f(\overline{z}) - f(\underline{z})]$ and $b = g(\underline{z}) - af(\underline{z})$. Finally, since $f(\overline{z}) > f(\underline{z})$, we have $g(\overline{z}) \geq g(\underline{z})$, and thus $a \geq 0$. *Q.E.D.*

Proof of Proposition 1. Let $A = \{x, y\}$. Since $(v+\tau)(x) - (v+\tau)(y) = s - k$, the option x maximizes $v + \tau$ over A if $k \leq s$, and y does if $k \geq s$. If $k < s$, then $\hat{x}(A) = x$ and $c(A) = \max\{\tau(x), \tau(y)\} - \tau(x) = \max\{k, 0\}$, which gives $Q(x, A) = \max\{k, 0\} < s$ and $Q(y, A) = \max\{k, 0\} - s$. If $k \geq s$, take $\hat{x}(A) = y$; since $k > 0$, $c(A) = \max\{\tau(x), \tau(y)\} - \tau(y) = 0$, which gives $Q(x, A) = s$ and $Q(y, A) = 0$. In both cases, $Q(x, A) = \min\{\max\{k, 0\}, s\}$ and $Q(y, A) = Q(x, A) - s$. *Q.E.D.*

Proof of Proposition 2. Suppose first that $\tau = av + \delta$. Then $v + \tau = (1 + a)v + \delta$. Fix $A \in \mathbb{M}$. If $a \geq 0$, both $v + \tau$ and τ attain their maxima over A at a most ideal option, and $R(A) = \max_{x \in A} v(x)$. If $-1 < a < 0$, then $\max_{y \in A}(v + \tau)(y) = (1 + a) \max_{x \in A} v(x) + \delta$ and $\max_{z \in A} \tau(z) = a \min_{x \in A} v(x) + \delta$, and $R(A) = (1+a) \max_{x \in A} v(x) - a \min_{x \in A} v(x)$. If $a \leq -1$, then $\max_{y \in A}(v + \tau)(y) = (1 + a) \min_{x \in A} v(x) + \delta$ and $\max_{z \in A} \tau(z) = a \min_{x \in A} v(x) + \delta$, and $R(A) = \min_{x \in A} v(x)$. In each case, the benchmark is a fixed fraction with the stated θ.

Conversely, suppose that the benchmark is a fixed fraction θ. Take $x, y \in X$ with $v(x) > v(y)$, and define s and k as in Section 3.1. By Proposition 1, $R(\{x, y\}) = v(x) - \min\{\max\{k, 0\}, s\}$, while the fixed fraction gives $R(\{x, y\}) = v(x) - (1 - \theta)s$. Hence

$$\min\{\max\{k, 0\}, s\} = (1 - \theta)s \qquad \text{whenever } v(x) > v(y). \tag{2}$$

If $0 < \theta < 1$, then (2) gives $k = (1 - \theta)s$, that is, $\tau(y) + (1 - \theta)v(y) = \tau(x) + (1 - \theta)v(x)$. The affine function $g := \tau + (1 - \theta)v$ therefore takes the same value at any two lotteries with different values under v. If $v(x) = v(y)$, pick $z \in X$ with $v(z) \neq v(x)$, which exists by assumption; then $g(x) = g(z) = g(y)$. Hence g takes the same value at every lottery, and $\tau = -(1 - \theta)v + \delta$ for some δ, with $a = \theta - 1 \in (-1, 0)$. If $\theta = 1$, then (2) gives $k \leq 0$, that is, $\tau(x) \geq \tau(y)$ whenever $v(x) > v(y)$. By Lemma 3, $\tau = av + \delta$ with $a \geq 0$. If $\theta = 0$, then (2) gives $k \geq s$, that is, $-(v + \tau)(x) \geq -(v + \tau)(y)$ whenever $v(x) > v(y)$. By Lemma 3, $-(v + \tau) = a'v + \delta'$ for some $a' \geq 0$ and $\delta' \in \mathbb{R}$, and $\tau = -(1 + a')v - \delta'$, with $a = -(1 + a') \leq -1$. *Q.E.D.*

Proof of the uniqueness part of Theorem 2. Let (u', v', τ') be another PDC representation of $\succsim$, with benchmark value R'. Both are PC representations, and Theorem 1 gives $u' = \alpha u + \beta_1$, $v' = \alpha v + \beta_2$, and $R' = \alpha R + \beta_2$ for some $\alpha > 0$ and $\beta_1, \beta_2 \in \mathbb{R}$.

(ii) implies (i). Suppose that (ii) holds. Then $v(x) > v(y)$ for some $x, y \in X$: otherwise $R(A)$ equals the common value of v for every A, and $Q(x, A) = 0$ for every choice. It remains to show that $\tau' = \alpha\tau + \beta_3$ for some $\beta_3 \in \mathbb{R}$.

Suppose first that $\tau \neq av + b$ for all $a, b \in \mathbb{R}$. Together with $v(x) > v(y)$ for some x, y, this is the regularity condition of Gul and Pesendorfer (2001), with v and τ in the roles of their commitment and temptation utilities. Both (v, τ) and (v', τ') represent $\succeq_R$, the binary relation over menus represented by R, in the form of their Theorem 1. This relation also has what they call self-control. Indeed, $v + \tau$ takes different values at some two lotteries, since $\tau \neq -v + b$. Moreover, $\tau \neq a(v + \tau) + b$ for all $a \geq 0$ and $b \in \mathbb{R}$: $a = 0$ would give $\tau = b$, $a = 1$ would give $v = -b$, and any other a would make τ of the form $a''v + b''$. The conclusion of Lemma 3 therefore fails for $f = v + \tau$ and $g = \tau$, and its hypothesis must fail as well: some $x, y \in X$ satisfy $(v + \tau)(x) > (v + \tau)(y)$ and $\tau(y) > \tau(x)$. For these lotteries, $R(\{x, y\}) = v(x) - [\tau(y) - \tau(x)]$, and hence $R(\{x\}) > R(\{x, y\}) > R(\{y\})$. Their Theorem 4 now yields $v' = \alpha' v + \beta_2'$ and $\tau' = \alpha'\tau + \beta_3$ for some $\alpha' > 0$ and $\beta_2', \beta_3 \in \mathbb{R}$. Since $v' = \alpha v + \beta_2$ and $v(x) > v(y)$ for some x, y, $\alpha' = \alpha$.

Suppose next that $\tau = av + b$ for some $a, b \in \mathbb{R}$. By Proposition 2, the benchmark is a fixed fraction θ. If $\theta = 1$, then $R(A) = \max_{x \in A} v(x)$ and $Q(x, A) \leq 0$ for every choice, which rules out the praise in (ii). If $\theta = 0$, then $R(A) = \min_{x \in A} v(x)$ and $Q(x, A) \geq 0$ for every choice, which rules out the blame in (ii). Hence $-1 < a < 0$ and $\theta = 1 + a \in (0, 1)$. Since $R' = \alpha R + \beta_2$ and $v' = \alpha v + \beta_2$ with $\alpha > 0$, R' is the same fixed fraction θ under v'. Proposition 2, applied to (u', v', τ'), gives $\tau' = av' + b'$ for some $b' \in \mathbb{R}$, and therefore $\tau' = \alpha\tau + \beta_3$ for some $\beta_3 \in \mathbb{R}$.

(i) implies (ii). Suppose that (ii) fails. If $v(x) = v(y)$ for all $x, y \in X$, then $R(A)$ equals the common value of v for every belief, and (u, v, τ'') is a PDC representation of $\succsim$ for every affine τ''. Let $\tau'' = -\tau$ if τ takes different values at some two lotteries, and let τ'' be any affine function that does otherwise. In both cases, $\tau'' \neq \alpha\tau + \beta_3$ for all $\alpha > 0$ and $\beta_3 \in \mathbb{R}$, and (i) fails.

If instead $v(x) > v(y)$ for some $x, y \in X$, then no choice earns praise, or no choice earns blame. In the first case, $Q(x, A) \leq 0$ for every choice, and taking x to be a most ideal option in A gives $R(A) \geq \max_{x \in A} v(x)$. Since the benchmark value lies between the least and the most ideal values, $R(A) = \max_{x \in A} v(x)$ for every A: the benchmark is the fixed fraction $\theta = 1$. By Proposition 2, $\tau = av + b$ with $a \geq 0$, and $\tau'' := \tau + v$ yields the same benchmark value. In the second case, the same argument gives $\theta = 0$ and $\tau = av + b$ with $a \leq -1$, and $\tau'' := \tau - v$ yields the same benchmark value. In both cases, (u, v, τ'') is a PDC representation of $\succsim$. If it had the form in (i), then $v = \alpha v + \beta_2$ would force $\alpha = 1$, and $\tau'' - \tau = \pm v$ would take the same value at every lottery, a contradiction. *Q.E.D.*

Proof of Proposition 3. For (a), let θ be the fixed fraction. Since $v(x^*) \geq \max_{x \in B} v(x)$ and

$v(x_*) < \min_{x\in B} v(x)$,

$$R(B \cup \{x^*\}) = \theta v(x^*) + (1-\theta)\min_{x\in B} v(x) \geq R(B),$$
$$R(B \cup \{x_*\}) = \theta \max_{x\in B} v(x) + (1-\theta)v(x_*) \leq R(B).$$

The first inequality is strict if $\theta > 0$ and $v(x^*) > \max_{x\in B} v(x)$, and the second is strict if $\theta < 1$. Since $Q(y, A) = v(y) - R(A)$, this gives the two inequalities for $y \in B$. Moreover, for every $y \in B$,

$$Q(x^*, B \cup \{x^*\}) = (1-\theta)\Big[v(x^*) - \min_{x\in B} v(x)\Big] \geq (1-\theta)\Big[\max_{x\in B} v(x) - \min_{x\in B} v(x)\Big] \geq Q(y, B),$$

where the last inequality holds because $Q(y, B) \leq \max_{x\in B} v(x) - R(B)$.

For (b), suppose first that $\tau(x^*) \geq \tau(y)$ for every $y \in B$. Then x^* maximizes both v and τ, and hence $v + \tau$, over $B \cup \{x^*\}$. Taking $\hat{x}(B \cup \{x^*\}) = x^*$ gives $c(B \cup \{x^*\}) = 0$ and $Q(y, B \cup \{x^*\}) = v(y) - v(x^*)$ for every $y \in B \cup \{x^*\}$. Next, suppose that $(v+\tau)(w) \leq \max_{y\in B}(v+\tau)(y)$ and $\tau(w) \leq \max_{y\in B} \tau(y)$. Then adding w to B changes neither $\max(v+\tau)$ nor $\max \tau$ over the menu. Hence $R(B \cup \{w\}) = R(B)$, and $Q(y, B \cup \{w\}) = Q(y, B)$ for every $y \in B$. *Q.E.D.*

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